CFD results report

Sample: 90° water bend · Job CF-SAMPLE · 13 Oct 2026

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Summary

We simulated steady water (liquid) flow through the pipe or fitting in your geometry (bend90.stl), with the inlet set to 4 m/s (a flow rate of 0.007832 m^3/s). The headline result is a static pressure drop of 5,294 Pa from inlet to outlet: the pressure the flow loses passing through the part. The calculation converged after 226 iterations, so the solution is numerically stable.

Convergence: CONVERGED

Judged against: residual targets p < 0.0001, other residuals < 0.001; stop when settled: ON (static pressure drop change < 0.5% over 200 iterations).

Converged at iteration 226: every residual met its target (p 0.0001, others 0.001).

Headline results

Static pressure drop (inlet – outlet)	5,294 Pa
Total pressure drop	5,294 Pa
Loss coefficient K (inlet velocity)	0.663
Flow rate	0.007832 m^3/s
Mean inlet velocity	4 m/s

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Job and inputs

Application	Hydraulics / Pipe Network
Flow type	Internal flow through the uploaded fluid volume
Fluid	Water (liquid) (ρ = 998 kg/m³, ν = 1e-06 m²/s)
Turbulence model	k-ω SST
Analysis package	Standard
Iterations run / budget	226 / 1500
Geometry file	bend90.stl

Boundary conditions

In	Velocity Inlet: 4 m/s
Out	Pressure Outlet: 0 Pa

Assumptions

- Steady state (simpleFoam, SIMPLE algorithm).
- Incompressible, isothermal (no heat transfer), single-phase Newtonian fluid.
- RANS turbulence (k-ω SST); near-wall flow resolved to y+ ≈ 1 with prism layers; walls smooth, no-slip.
- Inlet speed 4 m/s over the inlet area 0.001958 m² = 0.007832 m³/s.
- Internal flow through the uploaded fluid volume: 1 inlet, 1 outlet(s) at the outlet pressure, no-slip walls; cell size 4.99 mm, 2.49 mm at the walls (hydraulic diameter 49.89 mm).

Mesh and convergence

Mesh

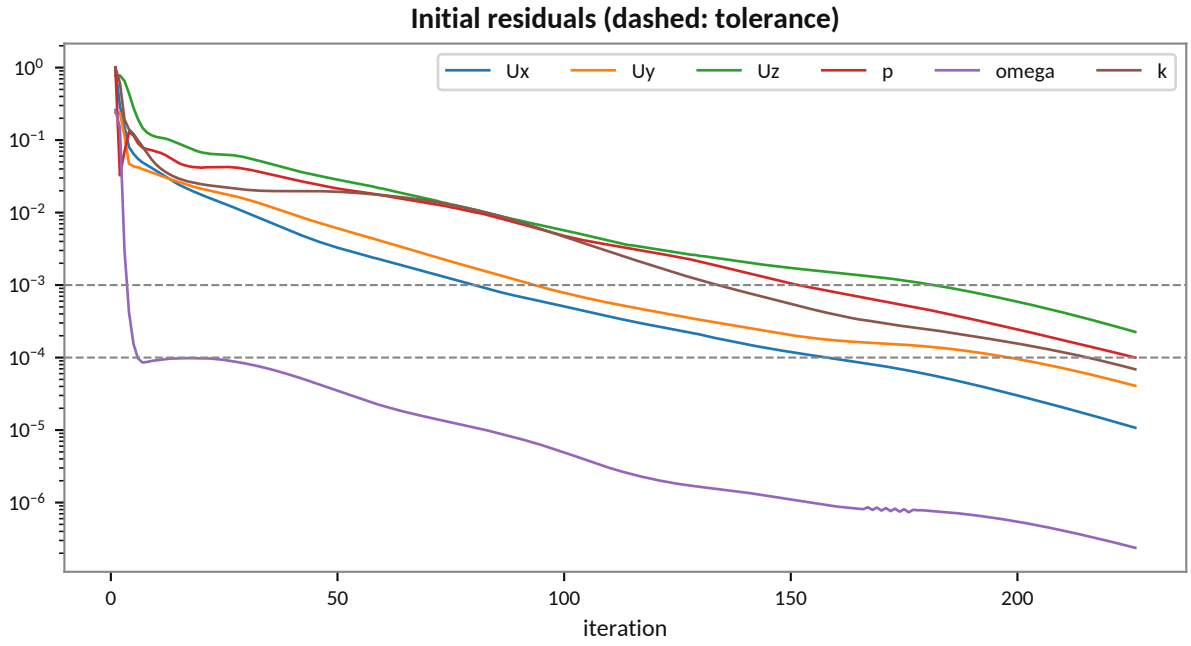
Cells	913,825
Wall layers	y+ ≈ 1 (resolved sublayer), 100 % of wall faces
Mesh checks (checkMesh)	passed
Max non-orthogonality	54.87°
Max skewness	0.6225
Negative volumes	none

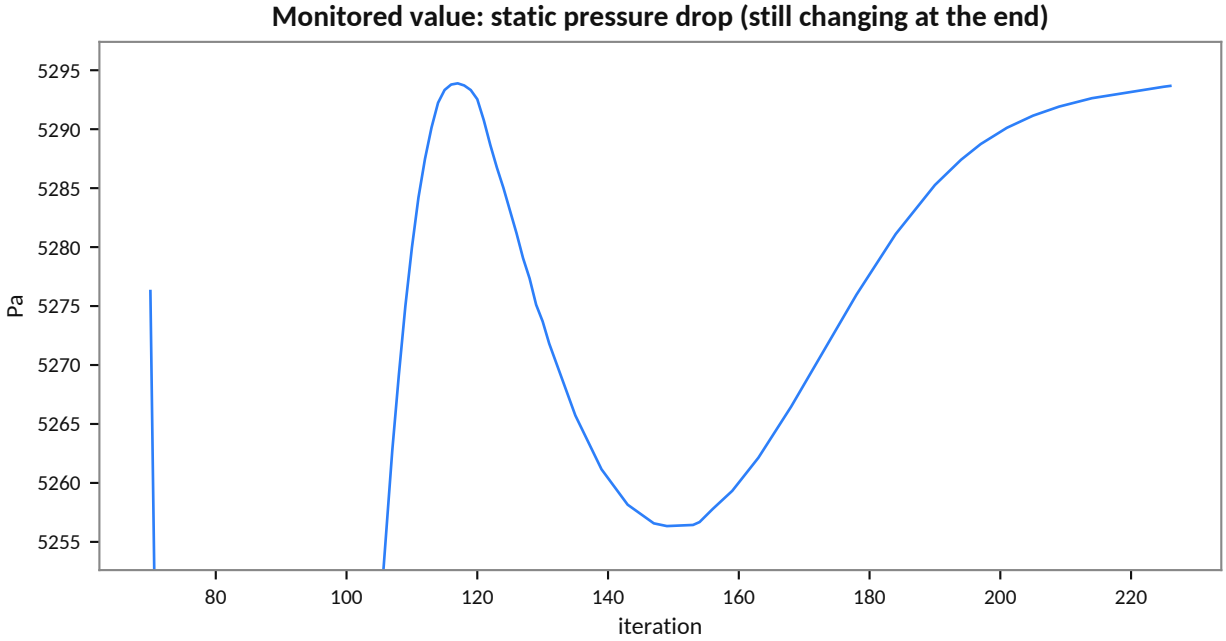
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y+ on walls

Walls	min 0.2735 · avg 1.042 · max 2.326

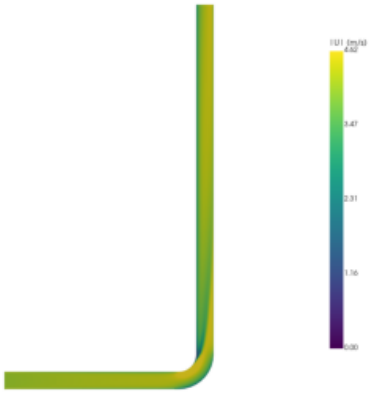
y+ ≈ 1: the viscous sublayer is resolved (low-Re wall treatment). y+ ≈ 30–300: wall functions model it. In between, the Spalding wall function blends the two.





Mid-plane fields

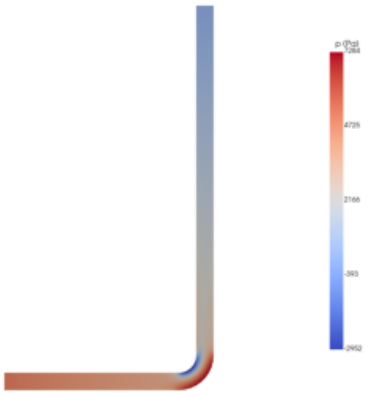
Velocity magnitude



Velocity, close-up of the fastest region



Static pressure (relative to the outlet)

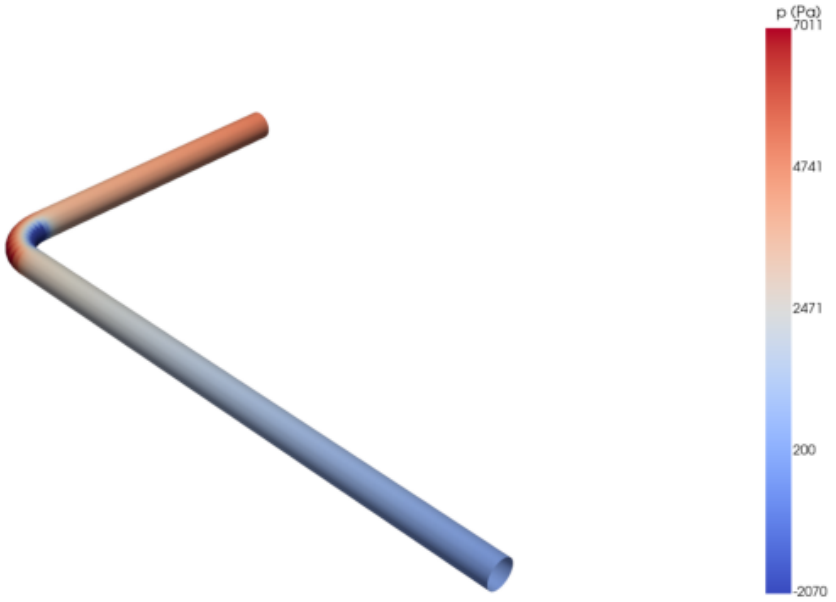


Pressure, close-up

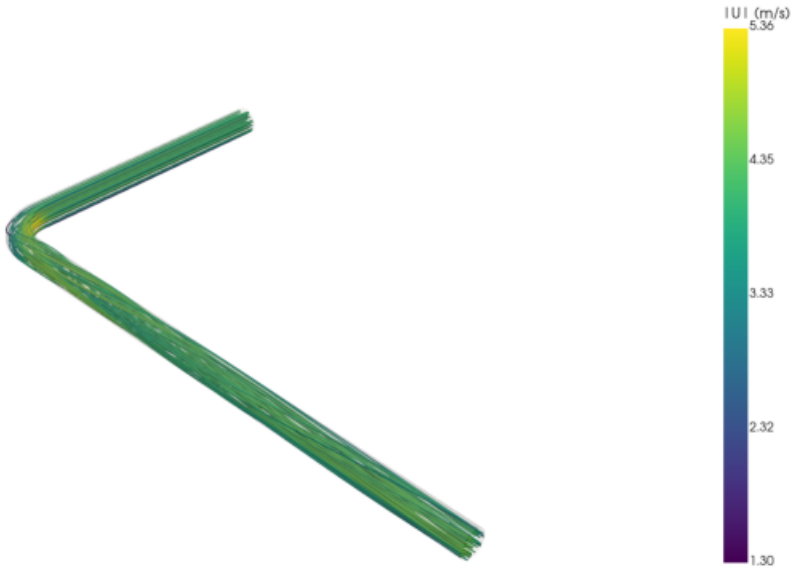


Surfaces and streamlines

Surface pressure



Streamlines coloured by speed



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